UNSIGNED INTEGER		OFFSET BINARY		SIGN AND MAGNITUDE		TWO'S COMPLEMENT	
Decimal	Bit Pattern	Decimal	Bit Pattern	Decimal	Bit Pattern	Decimal	Bit Pattern
15	1111	8	1111	7	0111	7	0111
14	1110	7	1110	6	0110	6	0110
13	1101	6	1101	5	0101	5	0101
12	1100	5	1100	4	0100	4	0100
11	1011	4	1011	3	0011	3	0011
10	1010	3	1010	2	0010	2	0010
9	1001	2	1001	1	0001	1	0001
8	1000	1	1000	0	0000	0	0000
7	0111	0	0111	0	1000	-1	1111
6	0110	-1	0110	-1	1001	-2	1110
5	0101	-2	0101	-2	1010	-3	1101
4	0100	-3	0100	-3	1011	-4	1100
3	0011	-4	0011	-4	1100	-5	1011
2	0010	-5	0010	-5	1101	-6	1010
1	0001	-6	0001	-6	1110	-7	1001
0	0000	-7	0000	-7	1111	-8	1000
16 bit range: 0 to 65,535		16 bit range -32,767 to 32,768		16 bit range -32,767 to 32,767		16 bit range -32,768 to 32,767	

FIGURE 4-1
Common formats for fixed point (integer) representation. Unsigned integer is a simple binary format, but cannot represent negative numbers. Offset binary and sign & magnitude allow negative numbers, but they are difficult to implement in hardware. Two's complement is the easiest to design hardware for, and is the most common format for general purpose computing.

These first three representations are conceptually simple, but difficult to implement in hardware. Remember, when A=B+C is entered into a computer program, some hardware engineer had to figure out how to make the bit pattern representing B, combine with the bit pattern representing C, to form the bit pattern representing A.

Two's complement is the format loved by hardware engineers, and is how integers are usually represented in computers. To understand the encoding pattern, look first at decimal number zero in Fig. 4-1, which corresponds to a binary zero, 0000. As we count upward, the decimal number is simply the binary equivalent (0 = 0000, 1 = 0001, 2 = 0010, 3 = 0011, etc.). Now, remember that these four bits are stored in a register consisting of 4 flip-flops. If we again start at 0000 and begin subtracting, the digital hardware automatically counts in two's complement: 0 = 0000, -1 = 1111, -2 = 1110, -3 = 1101, etc. This is analogous to the odometer in a new automobile. If driven forward, it changes: 00000, 00001, 00002, 00003, and so on. When driven backwards, the odometer changes: 00000, 99999, 99998, 99997, etc.

Using 16 bits, two's complement can represent numbers from -32,768 to 32,767. The left most bit is a 0 if the number is positive or zero, and a 1 if the number is negative. Consequently, the left most bit is called the **sign bit**, just as in sign & magnitude representation. Converting between decimal and two's complement is straightforward for positive numbers, a simple decimal to binary